Seniority-zero states are mean-field wavefunctions

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(Dated: August 15, 2025)

We use symbolic computation, numerical experiments, and mathematical arguments to establish that every seniority-zero state can be expressed as a (number-symmetry-broken) antisymmetric product of interacting geminals (APIG). Moreover, for bosons, the configuration interaction coefficients are exactly parameterized as matrix permanents. This establishes that bosonic wavefunctions can be exactly described by a mean field, specifically the (number-symmetry-broken) symmetric product of boson orbitals. While exact parameterization is extremely costly, we present numerical evidence showing that the APIG wavefunction approaches the true wavefunction exponentially quickly as the number of interacting geminals increases.

I. INTRODUCTION

In the 1960s, through the work of quantum chemistry luminaries like Thomas Allen, Harrison Shull, Darwin Smith, Enrico Clemente, Frank Weinhold, and E. Bright Wilson, it was recognized that a significant fraction of electron correlation in molecular systems can be modeled with a doubly-occupied configuration interaction (DOCI) wavefunction, in which every spatial orbital is either doubly-occupied or empty.[1–5] For a system with $N = 2M$ electrons in $Q = 2P$ spin-orbitals, DOCI wavefunctions have the form,

$$|\Psi_{\text{DOCI}}\rangle = \sum_{\left\{ m_j \in \{0,1\} \,\middle|\, M = \sum_{j=1}^{P} m_j \right\}} c^{\text{DOCI}}_{m_1 \ldots m_P} |\mathbf{m}\rangle, \tag{1}$$

where the (closed-shell) Slater determinants are represented by their occupation-number vectors (ONVs), typically encoded on computers as bitstrings,

$$|\mathbf{m}\rangle = \left(a_1^\dagger a_{\bar{1}}^\dagger\right)^{m_1} \left(a_2^\dagger a_{\bar{2}}^\dagger\right)^{m_2} \cdots \left(a_P^\dagger a_{\bar{P}}^\dagger\right)^{m_P} |\emptyset\rangle, \tag{2}$$

where $|\emptyset\rangle$ denotes the vacuum state. Typically $a_k^\dagger a_{\bar{k}}^\dagger$ generates an electron pair in the k-th spatial orbital, but it can be convenient to break (and optionally restore) spin symmetry:[6–9] the formulation only requires that a pairing scheme between disjoint sets of spin-orbitals $\{|k\rangle\}$ and $\{|\bar{k}\rangle\}$ be specified.

Over the last 50 years, the DOCI form has been repeatedly rediscovered and explored.[10–19] Arguably, the most significant recent advance was the link to the seniority quantum number, which specifies the number of unpaired electrons in a wavefunction.[15,16,20] From this perspective, DOCI is seniority-zero CI. This stimulated a new strategy for approaching the exact wavefunction, in which DOCI is extended by including terms with higher seniority, typically using configuration-interaction or coupled-cluster strategies.[16,21–30] However, the only size-consistent wavefunctions in the seniority hierarchy are seniority-zero (DOCI), full-seniority (no doubly-occupied orbitals), and all-seniority (full) CI. Approximations to DOCI, either by direct parameterization of the wavefunction or of its reduced density matrices, are typically size-consistent and often perform very well for problems like multiple-bond breaking, which are difficult to describe with traditional single-reference quantum chemistry methods.[23,31–47]

In 1969, David Silver showed how to approximate the DOCI wavefunction with an antisymmetric product of interacting geminals (APIG),[48,49]

$$|\Psi_{\text{APIG}}\rangle = \prod_{g=1}^{M} b_g^\dagger |\emptyset\rangle, \tag{3}$$

where

$$b_g^\dagger = \sum_{p=1}^{P} x_{gp} a_p^\dagger a_{\bar{p}}^\dagger. \tag{4}$$

In APIG, the DOCI coefficients are computed as the permanent of a matrix

$$\mathbf{X}_{\mathbf{m}} = \begin{bmatrix} x_{1n_1} & x_{1n_2} & \cdots & x_{1n_M} \\ x_{2n_1} & x_{2n_2} & \cdots & x_{2n_M} \\ \vdots & \vdots & \ddots & \vdots \\ x_{Mn_1} & x_{Mn_2} & \cdots & x_{Mn_M} \end{bmatrix}, \tag{5}$$

where one selects only the columns, indexed by $n_1 < n_2 < \cdots < n_M$, that correspond to occupied orbitals in $|\mathbf{m}\rangle$.[32,50] I.e.,

$$c^{\text{APIG}}_{\mathbf{m}}(\mathbf{X}) = \langle \mathbf{m} | \Psi_{\text{APIG}} \rangle = \text{per}(\mathbf{X}_{\mathbf{m}}), \tag{6}$$

or more explicitly,

$$c^{\text{APIG}}_{\mathbf{m}}(\mathbf{X}) = \text{per}(\mathbf{X}\mathbf{P}_{\mathbf{m}}), \tag{7}$$

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where $\mathbf{X}$ is the full matrix of geminal coefficients (cf. Eq. (4)) and the elements of $\mathbf{P_m}$ are defined as

$$p_{ij} = 0 \quad \text{unless} \quad m_j = 1 \quad \text{and} \quad j = 1 + \sum_{k=1}^{i-1} m_k. \tag{8}$$

$\mathbf{P_m}$ is a generalized permutation matrix corresponding to a specific way to select M objects from P choices. Its column sums are one and its row sums are zero or one.

For molecular systems, APIG,[48–53] and approximations to it like perfect-pairing generalized valence bond,[54–62] antisymmetric products of strongly orthogonal geminals,[54–56,63–68] antisymmetric products of rank-2 geminals,[69,70] antisymmetric products of 1-reference-orbital geminals (a.k.a. pair coupled-cluster doubles),[22,31,33,71] and Richardson-Gaudin states,[72–76] typically reproduce the energy of the DOCI wavefunction to within a few microHartree. This result is surprising because (a) APIG is just a product of electron pairs and, as such, is a mean-field wavefunction and (b) the number of free parameters in the DOCI wavefunction is far greater than the number of free parameters in APIG.

In the next two sections, we will define a hierarchy of three number-symmetry-broken APIG wavefunctions, followed by evidence from symbolic and numerical computation that number-symmetry-broken APIG provides an exact parameterization of DOCI. In particular, we present a robust numerical procedure that maps any DOCI state to its APIG form. Then, leveraging the fact that electron pairs are (hard-core) bosons, we generalize our results to bosonic wavefunctions. Because *every* bosonic state can be written as a number-symmetry-broken mean field, *any* bosonic wavefunction can be sampled by evaluating matrix permanents. Motivated by our numerical results, we present both a constructive procedure and a mathematical argument for the exactness of these wavefunction forms. No such construction works for fermions.

II. NUMBER-SYMMETRY-BROKEN ANTISYMMETRIC PRODUCTS OF INTERACTING GEMINALS (APIG)

The simplest number-symmetry-broken APIG wavefunction is

$$|\Psi_{\text{APIG1}}\rangle = \prod_{g=1}^{G} (1 + b_g^\dagger) |\emptyset\rangle, \tag{9}$$

where $b_g^\dagger$ is given by the APIG expression, Eq. (4). We allow the number of geminals G to exceed not only the number of pairs, M, but even the size of the basis, P. Projecting $|\Psi_{\text{APIG1}}\rangle$ onto the $2M$-electron Hilbert space produces a linear combination of APIG wavefunctions. The projection onto the the desired Hilbert space can be computed using, for example, the Peirls-Yoccoz projection,[77]

$$\hat{P}_{2M} = \frac{1}{2\pi} \int_0^{2\pi} \exp\left[i\phi \left(\sum_{p=1}^{P} a_p^\dagger a_p - 2M \right) \right] d\phi \tag{10}$$

Unlike arbitrary linear combinations of APIG wavefunctions, the product structure in Eq. (9) ensures that size consistency is retained. Number-symmetry broken expressions with the form (9) are common in nuclear and condensed-matter physics, but less common in quantum chemistry.[8,78] For example, the BCS wavefunction has a similar form, and it is known to be exact in the thermodynamic limit ($G \to \infty$) for certain systems.[79] Our key result is that there exists a $G < \infty$ for which Eq. (9) is exact for *any* seniority-zero state.

The CI coefficients of $|\Psi_{\text{APIG1}}\rangle$ can be computed by augmenting the matrix in Eq. (6) by $G-M$ columns of 1's. However, it is more convenient to express the CI coefficients as a *rectangular permanent*:

$$c_{\mathbf{m}}^{\text{APIG1}}(\mathbf{X}) = (G-M)!\text{per} \begin{bmatrix} x_{1n_1} & x_{1n_2} & \cdots & x_{1n_M} \\ x_{2n_1} & x_{2n_2} & \cdots & x_{2n_M} \\ \vdots & \vdots & \ddots & \vdots \\ x_{Gn_1} & x_{Gn_2} & \cdots & x_{Gn_M} \end{bmatrix}. \tag{11}$$

Here we have used the fact that the permanent of an $n \times n$ matrix of 1's is $n!$. Recall that a rectangular permanent is defined by summing over all possible ways to select M of the G rows, then using the selected rows to evaluate the permanent.[80–82]

Multiplying a matrix by a diagonal matrix changes its permanent by the product of the diagonal elements,

$$\text{per}(\mathbf{D} \cdot \mathbf{X_m}) = \text{per}(\mathbf{D})\,\text{per}(\mathbf{X_m}) = \left(\prod_{g=1}^{G} d_{gg} \right) \text{per}(\mathbf{X_m}). \tag{12}$$

This results in a slight generalization of the $|\Psi_{\text{APIG1}}\rangle$ form:

$$|\Psi_{\text{APIGa}}\rangle = \prod_{g=1}^{G} (x_{g0} + b_g^\dagger) |\emptyset\rangle. \tag{13}$$

Projecting the number-symmetry-broken form in Eq. (13) onto the $2M$-electron Hilbert space gives

$$c_{\mathbf{m}}^{\text{APIGa}}(\mathbf{X}) = \text{per} \begin{bmatrix} x_{1n_1} & x_{1n_2} & \cdots & x_{1n_M} & x_{10} & \cdots & x_{10} \\ x_{2n_1} & x_{2n_2} & \cdots & x_{2n_M} & x_{20} & \cdots & x_{20} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ x_{Gn_1} & x_{Gn_2} & \cdots & x_{Gn_M} & \underbrace{x_{G0} \quad \cdots \quad x_{G0}}_{G-M \text{ times}} \end{bmatrix}. \tag{14}$$

A somewhat more elaborate approach introduces B bath orbitals,

$$\begin{aligned} |\Psi_{\text{APIGb}}\rangle &= \prod_{g=1}^{G} b_g^\dagger |\emptyset\rangle \\ b_g^\dagger &= \sum_{p=1}^{P} x_{gp} a_p^\dagger a_{\bar{p}}^\dagger + \sum_{p=P+1}^{P+B} x_{gp} a_p^\dagger a_{\bar{p}}^\dagger. \end{aligned} \tag{15}$$

One then projects out the bath to construct the fragment wavefunction for the $2M$-electron system. This is simplest if one adds the minimal number of bath states, $B = G - M$. Then one possible way to construct a system with $2M$ electrons in the

system is to choose the bath to be full, which corresponds to appending a string of $G-M$ 1's to $\mathbf{m}$,

$$\mathbf{m}_{\text{sys+bath}} = \begin{bmatrix} m_1 & m_2 & \dots & m_P & 1 & 1 & \cdots & 1 \end{bmatrix}. \tag{16}$$

Projecting out the bath gives

$$\begin{aligned} c^{\text{APIGb}}_{\mathbf{m}_{\text{sys+bath}}}(\mathbf{X}) &= \left\langle \mathbf{m}_{\text{sys+bath}} \middle| \Psi_{\text{APIGb}} \right\rangle = \text{per}(\mathbf{X}\mathbf{P}_{\mathbf{m}_{\text{sys+bath}}}) \\ &= \text{per} \begin{bmatrix} x_{1n_1} & x_{1n_2} & \cdots & x_{1n_M} & x_{1n_{M+1}} & \cdots & x_{1n_G} \\ x_{2n_1} & x_{2n_2} & \cdots & x_{2n_M} & x_{2n_{M+1}} & \cdots & x_{2n_G} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ x_{Gn_1} & x_{Gn_2} & \cdots & x_{Gn_M} & x_{Gn_{M+1}} & \cdots & x_{Gn_G} \end{bmatrix}. \end{aligned} \tag{17}$$

where $\mathbf{P}_{\mathbf{m}}$ is constructed via Eq. (8). In this formulation, $|\Psi_{\text{APIGb}}\rangle$ is mathematically similar to traditional, non-symmetry-broken, APIG: the addition of the bath only changes the number of electrons and orbitals. The "bath trick" is closely related to the strategy used in particle-breaking Hartree-Fock theory-based embedding.[83–85]

III. SUFFICIENCY OF NUMBER-SYMMETRY BROKEN APIG

A. Symbolic Computation

Can number-symmetry broken APIG provide an exact parameterization of DOCI wavefunctions. In particular, for any DOCI wavefunction, does there exist a $G < \infty$ such $c^{\text{DOCI}}_{\mathbf{m}} = c^{\text{APIG1}}_{\mathbf{m}}$?

We tested this hypothesis using symbolic and numerical computation. While we experimented with all the symmetry-broken APIG forms mentioned above, we did the most thorough analysis for $|\Psi_{\text{APIG1}}\rangle$, which has the fewest free parameters. Our strategy is to treat

$$\text{per} \begin{bmatrix} x_{1n_1} & x_{1n_2} & \cdots & x_{1n_M} \\ x_{2n_1} & x_{2n_2} & \cdots & x_{2n_M} \\ \vdots & \vdots & \ddots & \vdots \\ x_{Gn_1} & x_{Gn_2} & \cdots & x_{Gn_M} \end{bmatrix} - c^{\text{DOCI}}_{\mathbf{m}} = 0 \tag{18}$$

as a system of polynomial equations to solve. (We dropped the $(G-M)!$ prefactor from Eq. (11) because it is merely a scaling factor for the geminal coefficients). For 2 electron pairs in 4 and 5 spatial orbitals ($M = 2; P = 4,5$) we were able to verify, symbolically, that there is always a solution for any choice of $c^{\text{DOCI}}_{\mathbf{m}}$. Specifically, we used `SageMath` to verify that solutions exist by (1) constructing an ideal from the above system of polynomial equations, (2) computing a Gröbner basis for the ideal, and (3) confirming the solution set was nonempty.[86] For 4 spatial orbitals, there are 6 DOCI coefficients and 7 free APIG variables, so traditional APIG is adequate. For 5 spatial orbitals, there are 10 DOCI coefficients but APIG has only 9 free variables; unsurprisingly, we observed that there was no solution until three geminals were used ($G = 3$; 15 free variables). Note that traditional, non-number-symmetry-broken, APIG has one less degree of freedom than one would expect, because multiplying the coefficient matrix by a diagonal matrix with determinant 1 does not affect the permanent. I.e., the number of independent dimensions in a traditional APIG wavefunction, with $G = M$, is $GP - 1$.

To treat larger systems, we generated random DOCI coefficients, which we then scaled and rounded to integers. We then used `HomotopyContinuation.jl` to perform regeneration homotopy continuation[87] and confirm that the DOCI wavefunction can be exactly parameterized by number-symmetry-broken APIG for 2 electron pairs (with $M = 6$ and $M = 7$) and 3 electron pairs (with $M = 5$).[88] In all cases, there were solutions as soon as the number of independent geminal coefficients exceeded the number of DOCI coefficients. I.e., for 2 pairs in 6 orbitals (15 DOCI coefficients) we needed 3 geminals (18 variables); for 2 pairs in 7 orbitals (21 DOCI coefficients) we needed 4 geminals (28 variables). For 3 pairs in 5 orbitals (10 DOCI coefficients), 3 geminals (14 variables) sufficed.

B. Numerical Experiments

To investigate larger systems, we numerically solved the system of polynomial equations,

$$c^{\text{APIG1}}_{\mathbf{m}}(\mathbf{X}) - c^{\text{DOCI}}_{\mathbf{m}} = 0. \tag{19}$$

The rectangular permanent is a polynomial where every monomial is multilinear (every x_{gp} appears raised to either the zeroth or first power) and has degree M:

$$c^{\text{APIG1}}_{\mathbf{m}}(\mathbf{X}) = \text{per}(\mathbf{X}_{\mathbf{m}}) = \sum_{1 \le g_1 < \cdots < g_M \le G} \sum_{\pi \in S_M} \prod_{j=1}^{M} x_{g_{\pi(j)} n_j}, \tag{20}$$

where n_j indexes the nonzero elements of $\mathbf{m}$ and the inner sum is over permutations of M elements. Because $c^{\text{APIG1}}_{\mathbf{m}}(\mathbf{X})$ is homogeneous of degree M,

$$\mathbf{X} \cdot \nabla c^{\text{APIG1}}_{\mathbf{m}}(\mathbf{X}) = M c^{\text{APIG1}}_{\mathbf{m}}(\mathbf{X}). \tag{21}$$

For the specific case of the (rectangular) permanent, one can understand this expression because one can perform a cofactor/Laplace expansion along the columns of the matrix; if one sums over all the columns of $\mathbf{X}$, then one has computed the cofactor expansion M times (once for each column). I.e.,

$$\begin{aligned} \text{per}(\mathbf{X}_{\mathbf{m}}) &= M^{-1} \sum_{i=1}^{M} \sum_{g=1}^{G} x_{gn_i} \text{per}(\mathbf{X}^{(-g,-i)}_{\mathbf{m}}) \\ &= M^{-1} \sum_{i=1}^{M} \sum_{g=1}^{G} x_{gn_i} \frac{\partial \, \text{per}(\mathbf{X}_{\mathbf{m}})}{\partial x_{gn_i}}, \end{aligned} \tag{22}$$

where $\mathbf{X}^{(-g,-i)}_{\mathbf{m}}$ denotes the minor formed by removing the g-th row and i-th column from $\mathbf{X}_{\mathbf{m}}$.

The system of polynomial equations to be solved can now be written as:

$$c^{\text{DOCI}}_{\mathbf{m}} = M^{-1} \sum_{i=1}^{M} \sum_{g=1}^{G} \text{per}(\mathbf{X}^{(-g,-i)}_{\mathbf{m}}) x_{gn_i}, \tag{23}$$

or, in matrix-vector form,

$$\mathbf{c}^{\mathrm{DOCI}} = M^{-1}\mathbf{J}(\mathbf{X})\cdot\mathbf{X} \tag{24}$$

where $\mathbf{c}^{\mathrm{DOCI}}$ is the vector of DOCI coefficients and the Jacobian, $\mathbf{J}(\mathbf{X})$, is a $\binom{P}{M}\times GP$ dimensional matrix. The elements of the j-th row of $\mathbf{J}$, corresponding to the CI coefficient $c^{\mathrm{APIG1}}_{\mathbf{m}_j}$, are zero except for elements x_{gn_i} where n_i refers to an occupied orbital in $\mathbf{m}_j$, in which case they are the corresponding cofactor, $\mathrm{per}(\mathbf{X}^{(-g,-i)}_{\mathbf{m_j}})$.

The system of polynomial equations (23) can be solved using a damped Newton method. Specifically, we solved the system,

$$\begin{aligned}\mathbf{J}\cdot\mathbf{\Delta}_k &= \mathbf{c}^{\mathrm{DOCI}} - \mathbf{c}^{\mathrm{APIG1}}(\mathbf{X}_k)\\ \mathbf{X}_{k+1} &= \mathbf{X}_k + \omega\mathbf{\Delta}_k\end{aligned} \tag{25}$$

where $0<\omega\leq 1$ is a damping parameter. In each step we iteratively reduced ω by a factor of two until the error in the reconstructed DOCI coefficients, $|\mathbf{c}^{\mathrm{DOCI}} - \mathbf{c}^{\mathrm{APIG1}}(\mathbf{X}_{k+1})|$, decreased from the previous iteration. In all but one of our tests, we could solve Eqs. (23) to a tolerance of 10^{-10}. In one test, denoted with a * in Table I, we converged to 5×10^{-10}. This is likely because the evaluation of the permanent is susceptible to round-off error, which accumulates *rapidly* with increasing matrix size.[89] We observed that increasing the number of unknown geminal parameters, G, facilitated convergence, perhaps because the Jacobian was further from singular (so larger steps were allowed) and because the additional dimensionality increases the likelihood of convergence. In our tests, using complex, instead of real, coefficients did not change our results qualitatively.

The most expensive step of this damped-Newton iteration is forming the Jacobian, which has $\binom{P}{M}$ rows, each of which requires the evaluation of GM permanents, which is costly because evaluating the permanent of a matrix is #P-complete.[90] To construct the permanents as efficiently as possible, we wrote a software package that heuristically selects an optimal algorithm for evaluating the permanent based on the dimensions of the (possibly rectangular) matrix.[91]

These iterative methods work well because singular Jacobians are rare. In the first instance, this follows from Sard's theorem: $\mathbf{c}(\mathbf{X})$ is a polynomial, which means that, except for possibly a set of measure zero, for any given $\mathbf{c}^{\mathrm{APIG1}}$ there exists an $\mathbf{X}$ that recovers that $\mathbf{c}$ for which the Jacobian has full row rank.[92,93] More intuitively, note that for a generic $\mathbf{X}$, every row of the Jacobian is nonzero on a distinct set of columns. For example, compare rows k and l. $j_{k;gp}=0$ for all p that are occupied in $\mathbf{m}_k$ but not in $\mathbf{m}_l$. If there are $2L$ indices for which $\mathbf{m}_k$ and $\mathbf{m}_l$ differ (i.e., the Hamming distance between the ONVs is $2L$), then there are LG places where $j_{k;gp}=0$ but $j_{l;gp}$ need not be, and LG places where $j_{l;gp}=0$ but $j_{k;gp}$ need not be. In general, then, the set of $\mathbf{X}$ for which $\mathbf{J}(\mathbf{X})=0$ is very small.

Given a $\mathbf{c}^{\mathrm{APIG1}}(\mathbf{X}_0)$ with a nonsingular Jacobian, the constant rank theorem implies that there is an open set with nonzero volume around $\mathbf{c}^{\mathrm{APIG1}}(\mathbf{X}_0)$ that can also be represented by symmetry-broken APIG.[93] Moreover, this set can be extended to infinity by diagonal scaling of the geminal matrix, $\mathbf{D}^{(1)}\mathbf{X}\mathbf{D}^{(2)}$. For example, the second diagonal factor means that for any $\mathbf{c}^{\mathrm{APIG1}}$ that can be represented by APIG, the DOCI coefficients $\prod_{j=1}^{P}\left(d^{(2)}_{jj}\right)^{m_j} c_{\mathbf{m}}$ are also representable. Extending this argument to *all* the points where $\mathbf{c}^{\mathrm{APIG1}}$ has a nonsingular Jacobian provides a strong plausibility argument for the sufficiency of the symmetry-broken APIG form, going beyond the simple dimensionality argument that a sufficiently underdetermined system of polynomial equations should have a solution almost everywhere. However, it does not seem easy to refine this argument into a formal proof for the sufficiency of the symmetry-broken APIG form. (That is, one needs to prove some additional property of the map $\mathbf{c}^{\mathrm{APIG1}}(\mathbf{X})$, e.g. that it is convex or that the set of $\mathbf{c}^{\mathrm{APIG1}}$ with singular Jacobians has co-dimension greater than one.)

In many real systems, adequate solutions can be found with fewer geminals than one would expect. To show this, we consider the DOCI wavefunction for BeH_2 at a few points in the

Table I. Tests for the numerical reconstruction of DOCI coefficients for M electron pairs in P spatial orbitals by a number-symmetry-broken APIG wavefunction with G geminals (cf. Eq.(9)). For each test, $n^{\mathrm{DOCI}}_{\mathrm{tests}}$ distinct sets of DOCI coefficients (with $|\mathbf{c}^{\mathrm{DOCI}}|^2=1$) were randomly generated. For every tested DOCI coefficient we started from $n^{\mathrm{APIG1}}_{\mathrm{tests}}$ random guesses for the geminal parameters, $\{x_{g,p}\}^{G,P}_{g=1,p=1}$, which were then scaled so that $|\mathbf{c}^{\mathrm{APIG1}}|^2=1$ before commencing the solution of the system of polynomial equations. All results were converged to within $\left|\mathbf{c}^{\mathrm{DOCI}}-\mathbf{c}^{\mathrm{APIG1}}(\mathbf{X})\right|<10^{-10}$ except in 1 of the 100 trials for the starred entry, where the error was 5×10^{-10}.

M	P	G	$n^{\mathrm{DOCI}}_{\mathrm{tests}}$	$n^{\mathrm{APIG1}}_{\mathrm{tests}}$
4	8	15	10	10
4	9	22	10	10
4	10	38	10	10*
5	10	34	3	3

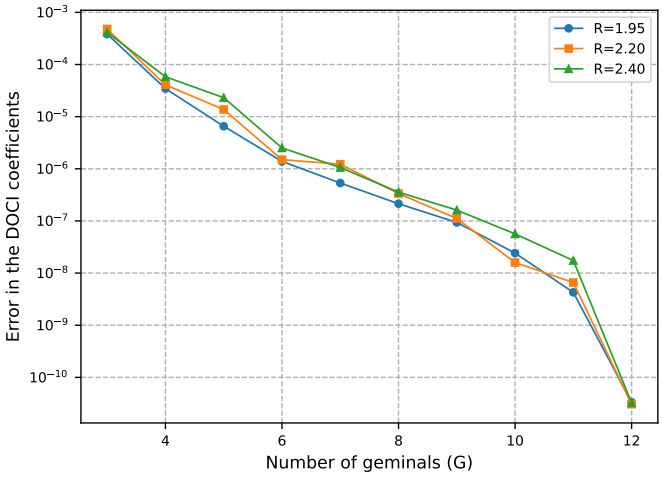

Figure 1. Error in the DOCI coefficients, $\left|\mathbf{c}^{\mathrm{DOCI}}-\mathbf{c}^{\mathrm{APIG1}}(\mathbf{X})\right|$, as a function of the number of geminals, G, for different separations, R, between Be and H_2. The Be atom was at the origin and the Hydrogen atoms are at $(0,\pm1.643,1.95);(0,\pm1.528,2.2);(0,\pm1.436,2.4)$ a.u..

transition-state region of the $\mathrm{Be}+\mathrm{H}_2$ dissociation reaction. We used the standard basis set of Purvis and Bartlett,[94,95] resulting in 3 electron pairs in 10 spatial orbitals, and solved for the orbital-optimized DOCI result using `PyCI`.[24,96] Figure 1 shows the coefficients of the DOCI wavefunction are well-produced with only a few interacting geminals, and that the error in the wavefunction coefficients decays exponentially as G increases. The sudden drop in error at $G=12$ occurs because the system of equations for the APIG coefficients is no longer overdetermined.

IV. BOSONS

While our primary interest is seniority-zero fermionic systems, the pair-creation operator, $a_k^\dagger a_{\bar{k}}^\dagger$ is equivalent to the creation operator for a hard-core boson in the state k, $\beta_k^\dagger$. This motivates consideration of the more general M-boson configuration interaction (bCI) wavefunction,

$$|\Psi_{\mathrm{bCI}}\rangle = \sum_{\left\{m_j\in\{0,1,\ldots,M\}\,\middle|\,M=\sum_{j=1}^P m_j\right\}} c^{\mathrm{bCI}}_{m_1\ldots m_P}|\mathbf{m}\rangle\,, \tag{26}$$

where the Slater permanents are represented by their ONVs,

$$|\mathbf{m}\rangle = \left(\beta_1^\dagger\right)^{m_1}\left(\beta_2^\dagger\right)^{m_2}\cdots\left(\beta_P^\dagger\right)^{m_P}|\emptyset\rangle\,, \tag{27}$$

and $\beta_k^\dagger$ adds a boson to the k-th orbital. Recently, there has been renewed interest in approximating the exact M-boson wavefunction with a boson mean field (BMF),[97–102]

$$|\Psi_{\mathrm{BMF}}\rangle = \prod_{g=1}^M b_g^\dagger|\emptyset\rangle \tag{28}$$

where

$$b_g^\dagger = \sum_{p=1}^P x_{gp}\beta_p^\dagger. \tag{29}$$

The bosonic CI coefficients associated with $|\Psi_{\mathrm{BMF}}\rangle$ are permanents,

$$\mathbf{c}^{\mathrm{BMF}}_{\mathbf{m}} = \frac{1}{\prod_{p=1}^P m_p!}\,\mathrm{per}\begin{bmatrix} x_{11}\cdots x_{11} & x_{12}\cdots x_{12} & \cdots & x_{1P}\cdots x_{1P}\\ x_{21}\cdots x_{21} & x_{22}\cdots x_{22} & \cdots & x_{2P}\cdots x_{2P}\\ \vdots & \vdots & \ddots & \vdots\\ \underbrace{x_{M1}\cdots x_{M1}}_{m_1\text{ times}} & \underbrace{x_{M2}\cdots x_{M2}}_{m_2\text{ times}} & \cdots & \underbrace{x_{MP}\cdots x_{MP}}_{m_P\text{ times}}\end{bmatrix} \tag{30}$$

We did not systematically explore symmetry-broken forms of the boson mean field wavefunction, e.g.

$$|\Psi_{\mathrm{BMFa}}\rangle = \prod_{g=1}^G (x_{g0}+b_g^\dagger)|\emptyset\rangle\,. \tag{31}$$

Note, however, that the equations to be considered are very similar to those associated with symmetry-broken APIG states. Moreover, M-boson CI can be rewritten in the DOCI form by duplicating each boson state M times,[103–105]

$$\begin{aligned}|\Psi_{\mathrm{bCI}}\rangle &= \sum_{\left\{m_{ij}\in\{0,1\}\,\middle|\,M=\sum_{i=1}^M\sum_{j=1}^P m_j\right\}} c^{\mathrm{bCI}}_{m_{11}\ldots m_{MP}}|\tilde{\mathbf{m}}\rangle\\ |\tilde{\mathbf{m}}\rangle &= \left(\beta_{11}^\dagger\right)^{m_{11}}\cdots\left(\beta_{1M}^\dagger\right)^{m_{1M}}\left(\beta_{21}^\dagger\right)^{m_{21}}\\ &\quad\cdots\left(\beta_{P-1,M}^\dagger\right)^{m_{P-1,M}}\left(\beta_{P1}^\dagger\right)^{m_{P1}}\left(\beta_{PM}^\dagger\right)^{m_{PM}}|\emptyset\rangle\,,\end{aligned} \tag{32}$$

Our previous tests on symmetry-broken APIG are directly applicable to this exact expression for the bosonic wavefunction.

Note that because determinants of matrices with repeated rows are zero, treating non-seniority-zero fermionic systems requires a different strategy.

V. THEOREMS

Inspired by our numerical results, we now state and prove the following theorem:

Theorem 1. *Any $2M$-electron DOCI wavefunctions (cf. Eq. (1)) can be generated by the projection of a number-symmetry-broken APIG wavefunctions (cf. Eq. (9)) onto the $2M$-electron Hilbert space.*

Proof. Start from the expansion of the M-pair DOCI wavefunction in Slater determinants,

$$|\Psi_{\mathrm{DOCI}}\rangle = \sum_{\left\{m_j\in\{0,1\}\,\middle|\,M=\sum_{j=1}^P m_j\right\}} c^{\mathrm{DOCI}}_{\mathbf{m}}\left(\beta_1^\dagger\right)^{m_1}\cdots\left(\beta_P^\dagger\right)^{m_P}|\emptyset\rangle\,, \tag{33}$$

where $\beta_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger$ creates an electron pair in the p-th orbital pair. Every Slater determinant can be written as an antisymmetrized geminal power (AGP),[106–108]

$$\left(\beta_1^\dagger\right)^{m_1}\cdots\left(\beta_P^\dagger\right)^{m_P}|\emptyset\rangle = \tfrac{1}{M!}\left(b_{\mathbf{m}}^\dagger\right)^M|\emptyset\rangle \tag{34}$$

where

$$b_{\mathbf{m}}^\dagger = \sum_{p=1}^P \delta_{m_p 1}a_p^\dagger a_{\bar{p}}^\dagger \tag{35}$$

sums over the occupied orbital pairs. So the DOCI wavefunction can be reexpressed as a linear combination of AGP wavefunctions,

$$|\Psi_{\mathrm{DOCI}}\rangle = \sum_{\left\{ m_j \in \{0,1\} \,\middle|\, M = \sum_{j=1}^{P} m_j \right\}} \frac{c_{\mathbf{m}}^{\mathrm{DOCI}}}{M!} \left(b_{\mathbf{m}}^{\dagger}\right)^{M} |\emptyset\rangle . \tag{36}$$

We now break number symmetry and restore it,

$$|\Psi_{\mathrm{DOCI}}\rangle = \hat{P}_{2M} \left[\prod_{\left\{ m_j \in \{0,1\} \,\middle|\, M = \sum_{j=1}^{P} m_j \right\}} \left(1 - \left(\omega_{\mathbf{m}} b_{\mathbf{m}}^{\dagger}\right)^{M}\right) |\emptyset\rangle \right], \tag{37}$$

where $\hat{P}_{2M}$ denotes projection onto the $2M$-electron Hilbert space and

$$\omega_{\mathbf{m}} = \left(-\frac{c_{\mathbf{m}}^{\mathrm{DOCI}}}{M!} \right)^{\frac{1}{M}} . \tag{38}$$

(Any M-th root can be selected.) Next, we invoke the identity

$$\left(1 - \left(\omega_{\mathbf{m}} b_{\mathbf{m}}^{\dagger}\right)^{M}\right) = \prod_{\ell=1}^{M} \left(1 - e^{\frac{2\ell\pi i}{M}} \omega_{\mathbf{m}} b_{\mathbf{m}}^{\dagger}\right), \tag{39}$$

which is often called a roots-of-unity filter. Inserting this expression into Eq. (37) gives an explicit expression for an arbitrary seniority-zero wavefunction in the $|\Psi_{\mathrm{APIG1}}\rangle$ form (cf. Eq. (9)):

$$|\Psi_{\mathrm{DOCI}}\rangle = \hat{P}_{2M} \left[\prod_{\left\{ m_j \in \{0,1\} \,\middle|\, M = \sum_{j=1}^{P} m_j \right\}} \prod_{\ell=1}^{M} \left(1 - e^{\frac{2\ell\pi i}{M}} \omega_{\mathbf{m}} b_{\mathbf{m}}^{\dagger}\right) |\emptyset\rangle \right]. \tag{40}$$

□

Note. *The mapping is clearly not unique. Even within this construction, which already chooses an exceptionally specific form for the geminals, one can use a different pairing scheme in Eq. (35), a different root of the CI coefficient in Eq. (38), or a different form of the roots-of-unity in Eq. (39).*

The result for a general M-boson wavefunction is very similar.

Theorem 2. *Any M-boson wavefunctions (cf. Eq. (26)) can be generated by the projection of a number-symmetry-broken boson product wavefunction (cf. Eq. (31)) onto the M-boson Hilbert space.*

Proof. We use the Waring decomposition of the M-boson wavefunction to reexpress it as a linear combination of boson powers,

$$\begin{aligned} |\Psi_{\mathrm{bCI}}\rangle &= \sum_{\left\{ m_j \in \{0,1,\ldots,M\} \,\middle|\, M = \sum_{j=1}^{P} m_j \right\}} c_{\mathbf{m}}^{\mathrm{bCI}} \left(\beta_1^{\dagger}\right)^{m_1} \cdots \left(\beta_P^{\dagger}\right)^{m_P} |\emptyset\rangle \\ &= \sum_{r=1}^{R} \left(b_r^{\dagger}\right)^{M} |\emptyset\rangle , \end{aligned} \tag{41}$$

where $b_r^{\dagger}$ has the form in Eq. (29). As before, we can explicitly construct $\sum_{r=1}^{R} \left(b_{r,\mathbf{m}}^{\dagger}\right)^{M}$ for each Slater permanent and then combine these forms,[109] but in general many fewer terms are required. For $P \geq 4$ and $M \geq 4$, $R = \binom{P+M-2}{M-1} - \binom{P+M-6}{M-3} - \binom{P+M-7}{M-4}$ suffices,[110,111] and, except for a set of measure, zero it even suffices to use $R = \left\lceil \frac{1}{P} \binom{P-1+M}{M} \right\rceil$.[112]

Replacing each boson power with the factorized expression,

$$\left(1 + (b_r^{\dagger})^{M}\right) = \prod_{\ell=1}^{M} \left(1 - e^{\frac{(2\ell+1)\pi i}{M}} b_r^{\dagger}\right), \tag{42}$$

and projecting onto M-boson terms gives the desired product-of-sums form:

$$|\Psi_{\mathrm{bCI}}\rangle = \hat{P}_{M} \left[\prod_{r=1}^{R} \prod_{\ell=1}^{M} \left(1 - e^{\frac{(2\ell+1)\pi i}{M}} b_r^{\dagger}\right) |\emptyset\rangle \right] \tag{43}$$

□

These constructive proofs use G that is far larger than what we used in our computational studies. To obtain a better idea of how large G needs to be in practice, recall that the space of wavefunctions for M bosons in P orbitals is isomorphic to the space of complex polynomials in P variables that are homogeneous of degree M, $\mathscr{P}_M$. (This is clear from Eq. (26) if you replace the boson creation operators with algebraic variables, $\{\beta_p\}_{p=1}^{P}$.)

Our goal, then, is to show that, for sufficiently large G, *any* polynomial in $\mathscr{P}_M$ is well-approximated by the homogeneous of degree M part of the polynomial,

$$p(\beta_1, \ldots, \beta_P) = \prod_{g=1}^{G} \left(1 + \sum_{p=1}^{P} x_{gp} \beta_p\right). \tag{44}$$

(For simplicity, we chose $x_{g0} = 1$ in Eq. (31).)

Using the generating function

$$F(t, \boldsymbol{\beta}) = \prod_{g=1}^{G} \left(1 + t \sum_{p=1}^{P} x_{gp} \beta_p\right), \tag{45}$$

the homogeneous of degree M part of $p(\boldsymbol{\beta})$ can be expressed as

$$p_M(\boldsymbol{\beta}) = \frac{1}{M!} \left[\frac{\partial^M F(t, \boldsymbol{\beta})}{\partial t^M} \right]_{t=0} \tag{46}$$

or, more explicitly,

$$p_M(\boldsymbol{\beta}) = \frac{1}{(G-M)!\prod_{p=1}^{P} m_p!}\,\mathrm{per}\begin{bmatrix} x_{10}\cdots x_{10} & x_{11}\beta_1\cdots x_{11}\beta_1 & \cdots & x_{1P}\beta_P\cdots x_{1P}\beta_P \\ x_{20}\cdots x_{20} & x_{21}\beta_1\cdots x_{21}\beta_1 & \cdots & x_{2P}\beta_P\cdots x_{2P}\beta_P \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{x_{G0}\cdots x_{G0}}_{G-M\text{ times}} & \underbrace{x_{G1}\beta_1\cdots x_{G1}\beta_1}_{m_1\text{ times}} & \cdots & \underbrace{x_{GP}\beta_P\cdots x_{GP}\beta_P}_{m_P\text{ times}} \end{bmatrix} \tag{47}$$

The dimension of $\mathscr{P}_M$ is $\binom{P+M-1}{M}$. We can then think of Eqs. (46) and (47) as different ways to express the map from the linear forms,

$$b_g(\boldsymbol{\beta}) = \sum_{p=1}^{P} x_{gp}\beta_p, \tag{48}$$

parameterized by the complex coefficients $\{x_{gp}\}_{g=1;p=1}^{G,P} \in \mathbb{C}$, to $\mathscr{P}_M$, which we denote

$$\Phi : \mathbb{C}^{GP} \to \mathscr{P}_M \tag{49}$$

Consider the Jacobian that explains how $p_M(\boldsymbol{\beta})$ changes when one changes a contributing linear factor,

$$\begin{aligned} dp_M(\boldsymbol{\beta}) &= J_h(\boldsymbol{\beta})db_h(\boldsymbol{\beta}) \\ J_h(\boldsymbol{\beta}) &= \frac{\partial p_M}{\partial b_h} = \hat{P}_{M-1}\left[\prod_{\substack{g=1\\g\neq h}}^{G}(1+b_g(\boldsymbol{\beta}))\right] \\ \frac{\partial p_M}{\partial x_{hp}} &= \frac{\partial p_M}{\partial b_h}\frac{\partial b_h}{\partial x_{hp}} = J_h(\boldsymbol{\beta})\beta_p \end{aligned} \tag{50}$$

Here P_{M-1} extracts the component of the polynomial that is homogeneous of degree $M-1$. $J_h(\boldsymbol{\beta})$ is a polynomial that is homogeneous of degree $M-1$, and if the span of $\{J_h\}_{g=1}^{G}$ is equal to $\mathscr{P}_{M-1}$, then the Jacobian, $\frac{\partial p_M}{\partial x_{gp}}$ has full row rank. We are then assured that, given $p_M(\boldsymbol{\beta})$, any other nearby $p_M(\boldsymbol{\beta})$ can be achieved by making a small change in one or more of the linear factors, $\left\{b_g(\boldsymbol{\beta})\right\}_{g=1}^{G}$.

Choose a random set of linear factors, $\left\{b_g^{(0)}(\boldsymbol{\beta})\right\}_{g=1}^{G}$ with $G \geq \binom{P+M-2}{M-1}$ and construct the associated homogeneous-of-degree-M polynomial, $p_M^{(0)}(\boldsymbol{\beta})$ with coefficients $c_{\mathbf{m}}^{(0)}$. With probability one, the Jacobian has full row rank. Consequently, there exists a Euclidean-open neighborhood of polynomial coefficients around $c_{\mathbf{m}}^{(0)}$ with nonzero volume that can also be constructed by a product of linear factors. However, this implies that the set of polynomials that can be constructed by a product of linear factors is dense in the Zariski topology.[113,114] Invoking Chevalley's theorem, the image of the polynomial map, Eq. (49), is also constructible, and a dense constructible subset of an irreducible space (here, $\mathscr{P}_M$) contains a nonempty Zariski-open subset. Zariski-open subsets are dense in the Euclidean topology. This ensures that the coefficients of any $p_M(\boldsymbol{\beta}) \in \mathscr{P}_M$ can be approximated arbitrarily closely by a product of linear factors.

While $G \geq \binom{P+M-2}{M-1}$ suffices for almost all $p_M(\boldsymbol{\beta})$, one can often use fewer coefficients. Merely ensuring that the number of free parameters, GP, exceeds the number of free coefficients in $p_M(\boldsymbol{\beta})$ gives a lower bound on G, so the required value of G lies in the interval,

$$\left\lceil \frac{1}{P}\binom{P+M-1}{M} \right\rceil \leq G_{\mathrm{BMF}} \leq \binom{P+M-2}{M-1}, \tag{51}$$

except on a set of measure zero.

The argument can be extended to the seniority-zero case: in that case one considers multilinear polynomials that are homogeneous of degree M. The Jacobian has full rank (with probability one), ensuring that the target coefficients can be reproduced arbitrarily accurately when $G_{\mathrm{APIG1}} \geq \binom{P}{M-1}$. This can be tightened to $G_{\mathrm{APIG1}} \geq \left\lceil \frac{1}{M}\binom{P}{M-1} \right\rceil$, but the argument is more complicated. (The essence is that one doesn't need to span the full space of homogeneous multilinear polynomials of degree $M-1$ because, in general, multiplying one of those polynomials by another linear factor gives a polynomial that is homogeneous of degree M but not multilinear. This reduces the dimension of the space of $(M-1)$-degree homogeneous multilinear polynomials one needs to span by a factor of M.) As a result, the G that is required lies in the interval,

$$\left\lceil \frac{1}{P}\binom{P}{M} \right\rceil \leq G \leq \binom{P}{M-1}, \tag{52}$$

except on a set of measure zero.

VI. CONCLUSION

Using formal mathematical arguments, symbolic computation, computational experiments, and an explicit construction, we show that *every* seniority-zero fermionic wavefunction and every bosonic wavefunction that is an eigenfunction of the particle number can be obtained by projecting a symmetry-broken mean field onto states with the target particle number. Specifically, projecting the number-symmetry-broken boson mean field wavefunction,

$$\begin{aligned} |\Psi\rangle &= \hat{P}_M \prod_{g=1}^{G}\left(1+b_g^\dagger\right)|\emptyset\rangle \\ b_g^\dagger &= \sum_{p=1}^{P} x_{gp}\beta_p^\dagger, \end{aligned} \tag{53}$$

onto M boson states is exact for sufficiently large G, which will always lie within the interval (51). When $\beta_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger$,

this means that doubly-occupied (seniority-zero) CI wavefunctions can be represented exactly with number-symmetry-broken antisymmetric products of interacting geminals. Even when G is not large enough to reproduce the DOCI wavefunction exactly, the error is small: our results suggest that the error in the APIG parameterization of the DOCI coefficients decays exponentially as the number of geminals increases.

Our result provides strong motivation for approaches to the quantum many-body problem based on breaking particle-number symmetry and, because it is based on overlaps of the wavefunction parameterization with Slater determinants, fits directly into the Flexible Ansätze for N-particle Configuration Interaction (FANCI) strategy some of us are developing.[44] It also provides a new perspective on methods which add higher-seniority corrections to a seniority-zero wavefunction:[22,67,115–125] because any seniority-zero state can be expressed as a mean field, all such methods can be reinterpreted as instances of the traditional "mean-field plus correction" strategy for modeling electron correlation in many-fermion quantum systems.

There may also be implications for photonic quantum computing. Theorem 2 shows how to parameterize *any* M-boson wavefunction using the permanent of a matrix. While this matrix is not unitary, it can always be dilated to a unitary matrix,[126–128] thereby providing an explicit pathway for sampling arbitrary boson wavefunctions. Unfortunately, our systematic construction of the matrix permanent that represents an arbitrary boson state uses the Waring decomposition of that state, which is NP-hard to compute.[129,130]

ACKNOWLEDGMENTS

The authors thank NSERC (PWA, PAJ, SDB), the Canada Research Chairs (PWA, SDB), and the Digital Research Alliance of Canada for financial support and computational resources. SDB and PWA acknowledge support from an NSERC Alliance Society grant.

CONFLICT OF INTEREST

The authors have no conflicts to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the authors upon reasonable request.

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